Problem 00000001082: the full-dimension MRD boundary

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Statement and scope

The printed conjecture gives, for fixed (m, n, k), the number of MRD-code equivalence classes as

$$q^{(k-1)(n-k+1)}\big(1 + O(q^{-1})\big).$$

Use the standard convention that a linear rank-metric code is an $\mathbb{F}_{q^m}$-linear subspace of $\mathbb{F}_{q^m}^n$ and k is its dimension over that extension field. Its dimension over $\mathbb{F}_q$ is mk, not k. We take

$$(m, n, k) = (3, 2, 2).$$

The source prints no restriction $k < n$ or minimum distance greater than one. This choice also satisfies the strict $n < m$ in Morrison's Definition II.11 [1]. The formal result concerns the asserted count of MRD codes in this standard linear model. It makes no separate Gabidulin-only classification claim and does not replace k by a matrix-space dimension.

It is enough to let $q = p$ range over all primes: an asymptotic formula for all prime powers must hold on this unbounded subfamily. We will derive the exact count 1 for every such p, and prove that it cannot have the displayed main term p and relative error $O(p^{-1})$.

The actual code and its distance

Fix any prime p. Let $K = \mathbb{F}_p$, let $L = \mathbb{F}_{p^3}$, and put $V = L^2$. The Lean construction uses the native field `GaloisField p 3`, a splitting field over `ZMod p`; its actual extension degree is 3 and its cardinality is p^3. For $v = (v_1, v_2) \in V$ define the standard rank weight and distance by

$$\mathrm{wt}(v) = \dim_K \langle v_1, v_2 \rangle_K, \qquad d_R(x, y) = \mathrm{wt}(x - y).$$

This is Morrison's Definition II.8 and Sheekey's vector model [1, 2]. For a code C the minimum is taken over *all* pairs $x, y \in C$ with $x \neq y$.

Every nonzero vector has positive rank weight: if its coordinate span had dimension zero, that span would be zero and both coordinates would vanish. The actual vector $u = (1, 0)$ has coordinate span $K \cdot 1$ of dimension one. Hence the full code $C = V$ has minimum distance exactly 1, attained at $(u, 0)$. Its distance set is nonempty; no empty-infimum convention is used. Its actual size is

$$|C| = |L|^{\dim_L C} = (p^3)^2 = p^6 = p^{\,3(2 - d_R(C) + 1)}.$$

This is precisely equality in the rank-metric Singleton bound, so C is MRD [2]. Both the minimum and the cardinality are calculated from the actual code, rather than inserted as its defining values.

The complete equivalence-class count

Let D be *any* L-linear code of length 2 and dimension 2. Since $D \subseteq V$ and $\dim_L D = \dim_L V = 2$, finite-dimensional linear algebra gives $D = V$. Thus the complete universe of $[2, 2]$ linear MRD codes consists of the single actual subspace V.

We use the standard linear rank-equivalence from Morrison's Definition III.4: D and E are equivalent if an invertible L-linear map $f : V \to V$ preserves the actual rank weight of every vector and maps D onto E. Identity, inverse and composition verify that this is an equivalence relation. The Lean quotient uses this relation, including the actual image-of-code condition. It is nonempty and a subsingleton, so its finite cardinality is exactly

$$A(p) = 1.$$

Uniqueness was proved before forming the quotient. Consequently allowing additional standard semilinear equivalences cannot change this one-class count; no arbitrary relation or chosen list of representative examples is being used to obtain the result.

Failure of the claimed asymptotic

For the fixed parameters, the printed exponent is $(2-1)(2-2+1) = 1$. The true relative error is therefore

$$R(p) = \frac{A(p)}{p} - 1 = \frac{1}{p} - 1, \qquad |R(p)| = \frac{p-1}{p}.$$

For every real constant B and every integer cutoff N, Euclid's theorem provides a prime $p > \max\{N, B+1\}$. At that actual prime,

$$|R(p)| = \frac{p-1}{p} > \frac{B}{p}.$$

Thus $R(p)$ is not $O(p^{-1})$, with all constants and eventual cutoffs quantified. More explicitly, if $A(p) = p(1+\epsilon(p))$ eventually and $\epsilon(p) = O(p^{-1})$, division by the nonzero p forces $\epsilon(p) = R(p)$ eventually, contradicting the preceding inequality. The formal proof uses native `Asymptotics.IsBigO` on the ordered type of all primes with its `atTop` filter, as well as the explicit arbitrarily-large-prime witnesses.

This disproves the unrestricted asymptotic clause as printed. It makes no assertion about variants excluding $k = n$, nontrivial minimum-distance families, or the unspecified cell-decomposition clause.

Formalization and reproducibility

The complete proof is in `lean/Main.lean`, namespace `TLMC1082`. Its principal declarations are:

Bridge	Declaration
Actual finite-field parameters	`actual_parameters`
True minimum and attainment	`full_minimumDistance`
Actual Singleton equality	`full_isMRD`
Entire code classification	`code_eq_top`
Standard equivalence quotient	`classCount_one`
All constants and cutoffs fail	`arbitrarily_large_prime_failure`
Native asymptotic impossibility	`printed_asymptotic_false`
Main result	`complete_counterexample`

The project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`.

First run `lean -version`; then `lean/verify.sh` runs the root build, warning-as-error namespace axioms audit and official full fresh-kernel replay. Every owned declaration, including generated names, is enumerated and checked with ordinary `#print axioms`. The allowed foundational axioms are only `propext`, `Classical.choice`, and `Quot.sound`.

References

[1] K. Morrison, *Equivalence for Rank-metric and Matrix Codes and Automorphism Groups of Gabidulin Codes*, arXiv:1304.0501v1, Definitions II.8, II.11 and III.4, printed pp. 4, 5 and 8. `https://arxiv.org/abs/1304.0501v1`.

[2] J. Sheekey, *MRD Codes: Constructions and Connections*, arXiv:1904.05813v1 (2019), Sections 1.1–1.2 and Definition 3, PDF pp. 2–3. `https://arxiv.org/abs/1904.05813v1`.